

Akshit Tyagi

Founding Member of Technical Staff @ Axiom — Foundation Models for Biology & Health

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WORK EXPERIENCE

JAN 2025 - PRESENT

Founding Member of Technical Staff at AXIOM

Multimodal foundation models for cellular phenotyping and drug safety

- Built *in silico* clinical risk models predicting drug-induced liver injury (DILI) from molecular structure (SMILES) — **0.92 AUC** (77% sensitivity, 90% specificity) on FDA DILIRank, matching or outperforming industry-standard 2D and 3D spheroid *in vitro* assays
- Co-designed the underlying 2D multiplexed cytotoxicity assay with biologists and external research labs / CROs, using model interpretability to guide what the assay should measure
- Owned the imaging ML stack end to end: deep-learning segmentation and phenotype quantification, multimodal LLMs for mechanism-of-action classification, and pipelines over 10M+ cell images

JULY 2023 - 2024

Research Engineer at HELM AI

Using deep teaching for foundation models in Autonomous Vehicles

- Trained and fine-tuned UNets, Latent Diffusion Models and DINO-family models for perception and generative simulation, building physics-informed dash-cam video to augment real-world data
- Built deep-teaching methods for semi-supervised learning of self-driving foundation models; optimized the self-driving SDK for custom architectures, improving on-device latency

MAY 2022 - JULY 2023

Research Engineer at GOOGLE DEEPMIND

Google Health: clinical ML in research collaboration with Stanford hospitals

- Research collaboration with clinicians and researchers at Stanford hospitals on EHR-based patient risk prediction
- Built algorithms to infer patient medical condition from longitudinal EHR data using T5 transformers
- Ran the resulting models in shadow deployment at Stanford against live clinical data
- Surfaced instability in clinical risk-stratification models under distribution shift; published at NeurIPS 22 (ML4H)

NOV 2021 - MAR 2022

ML Engineer at AKASA — *multi-modal, context-aware representations of care-provider/patient interactions from insurance cards and prior medical history*

MAR - OCT 2021

ML Engineer at ARMORBLOX

Building and improving email security using NLP

- Built time-series anomaly-detection and out-of-distribution models over composite features from users' email profiles, for account-compromise detection
- Applied weak supervision to cut model maintenance under distribution drift

JULY 2020 - JAN 2021

AI Resident at X, THE MOONSHOT FACTORY — *ultra-low-power TinyML for iyo*

PUBLICATIONS AND POSTERS

Instability in clinical risk stratification models using deep learning. NeurIPS 22 (ML4H)

Daniel Lopez-Martinez, Alex Yakubovich, Martin Seneviratne, Adam D Lelkes, Akshit Tyagi, Jonas Kemp, Ethan Steinberg, Lance Downing, Ron Li, Keith Morse, Nigam Shah, Ming-Jun Chen.

Fast Intent Classification for Spoken Language Understanding. ICASSP 2020

Akshit Tyagi, Varun Sharma, Rahul Gupta, Lynn Samson, Nan Zhuang, Zihang Wang, Bill Campbell.

Conversational Question Answering and Noise Robustness. Amazon Research Days 2019

Akshit Tyagi, Anjishnu Kumar, Abhinav Sethy.

SELECTED RESEARCH & PROJECTS

AXIOM **DILI Assay Development & Structure-to-Risk Modeling** — *presented at SoT 2025*

Co-designed the 2D multiplexed cytotoxicity assay with biologists and external labs / CROs — cell-painting imaging plus LDH and MT-Glo readouts across >100,000 molecules profiled in primary human hepatocytes — closing the loop by feeding assay output back through the vision model to surface which phenotypes carry biological signal. Trained models to predict dose-dependent responses for >10 distinct cell stress and death phenotypes directly from SMILES. Fine-tuned a multimodal LLM on cell images together with ADME properties and human exposure data to predict dose-dependent response in-circuit, yielding a clinical DILI risk score benchmarked against 2D HepG2, 2D PHH imaging and 3D spheroid assays.

AXIOM **Cellular Imaging ML Stack**

Built and operate the imaging stack behind the assay work: microscope-level correction so instrument variation doesn't propagate downstream; a pipeline that auto-ingests and normalises each new batch into a growing 10M+ image corpus, with drift monitoring that alerts the team when incoming data or retrained models move out of tolerance; deep-learning segmentation quantifying vacuoles, cytoplasm, mitochondria and RNA condensates across cell-painting channels, featurized into the risk model to map imaged DILI presentation to risk; and fine-tuned DINO embeddings clustering phenotypes to surface bioactive molecules distinguishable from background perturbation.

GOOGLE DEEPMIND **Clinical Risk Models with Stanford Hospitals** — *NeurIPS 22 (ML4H)*

Showed that risk-stratification models with near-identical aggregate performance can disagree substantially on individual patients — an instability that undermines clinical trust, and one that has since grown into its own line of research. Built T5-based models over longitudinal EHR data with clinicians at Stanford, studied approaches to improve reliability, and validated in shadow deployment against live clinical data.

AMAZON / UMASS **Fast Intent Classification for Spoken Language Understanding** — *ICASSP 2020*

Early work on adaptive-depth inference — a technique now standard in efficient LLM serving and on-device speech systems. Applied a BranchyNet early-exit scheme to intent classification, adding exit points at intermediate stages so simpler utterances egress early rather than running the full network — cutting computational cost without compromising accuracy (~20% reduction in effective model size) on latency-constrained devices.

TECHNICAL SKILLS

METHODS: LLM fine-tuning & alignment (LoRA, RLHF, DPO); diffusion models; vision foundation models (DINO); segmentation; multimodal modeling

SCALE & DATA: Distributed data pipelines at 10M+ image scale (Ray, Spark, Modal)

LANGUAGES: Python, C++, CUDA

ML FRAMEWORKS: PyTorch, Hugging Face (Transformers, Datasets)

INFRA / MLOPS: Weights & Biases, MLflow, Docker, GCP, AWS

EDUCATION & AWARDS

M.S. Computer Science, UMass Amherst (2020) · **B.Tech. Electrical Engineering**, IIT Delhi (2018)

Awards: DISA, IIT Delhi (2015) · Institute Award, top 7% first year · KVPY Fellowship (2013)

Service: Reviewer — NeurIPS, ICML, ICLR, ACL